

QK RESEARCHER - INTERNSHIP PROJECT REPORT

Autonomous Financial Intelligence & Statutory 10-K Lineage Verification

Candidate / Author Name:	Quaid Khan
Project Name:	ARA-1 / QK Researcher (Financial Agent)
GitHub Repository:	https://github.com/Quaid-khan/financial-research-agent
Live Web Application:	https://huggingface.co/spaces/QuAiD11/qk-researcher
Verification Status:	100% Verified (64/64 Pytest Suite Passed)
Date of Submission:	August 2026

1. Executive Summary

QK Researcher (ARA-1) is an institutional-grade autonomous financial intelligence agent designed to address the critical challenge of financial hallucination and source restatement ambiguity in LLM-assisted equity research. Unlike basic search tools or chatbot wrappers, QK Researcher enforces strict 3-year statutory primary Form 10-K filing binding, period lineage quality gates, multi-market regulatory routing (US SEC, Pakistan PSX/SBP, UK Companies House), 5-year DCF intrinsic valuation modeling, vector memory recall over ChromaDB, and publication-grade PDF synthesis.

2. System Architecture & Key Components

- **Autonomous ReAct Orchestrator:** Executes Plan-and-Execute reasoning loops with Step 0 ChromaDB vector memory recall, circuit breaker fault-tolerance, and multi-hop fallback routing.
- **Primary Filing Binding Engine:** Dynamically queries SEC EDGAR company facts and binds target fiscal years (FY2024, FY2023, FY2022) to original statutory Form 10-K filings with zero comparative restatement substitution.
- **Multi-Market Regulatory Router:** Supports universal public companies across US SEC EDGAR (SECProvider), Pakistan Stock Exchange (PSXProvider), State Bank of Pakistan (SBPPProvider), and international markets.
- **Financial Calculation & DCF Engine:** Performs safe arithmetic, profitability margins (Gross, Operating, Net), ROE, ROA, CAGR, and 5-Year Discounted Cash Flow (DCF) valuation.
- **Long-Term Vector Memory:** Stores 384-dimensional semantic embeddings (all-MiniLM-L6-v2) in persistent local ChromaDB storage for instant recall.
- **Live Top Market Movers Ticker:** Auto-scrolling marquee bar tracking live prices and 24h % changes for top 10 market movers with 30s background TTL caching.

3. Primary Filing Lineage Audit Verification Results

Company	Fiscal Year	Revenues	Net Income	Statutory Accession No.	Filing Date	Status
JPMorgan (JPM)	FY2024	\$177.556B	\$58.471B	0000019617-25-000270	2025-02-14	Primary 10-K Valid
JPMorgan (JPM)	FY2023	\$158.104B	\$49.552B	0000019617-24-000225	2024-02-16	Primary 10-K Valid
JPMorgan (JPM)	FY2022	\$128.695B	\$37.676B	0000019617-23-000231	2023-02-21	Primary 10-K Valid
Apple (AAPL)	FY2024	\$391.035B	\$93.736B	0000320193-24-000123	2024-11-01	Primary 10-K Valid
Apple (AAPL)	FY2023	\$383.285B	\$96.995B	0000320193-23-000106	2023-11-03	Primary 10-K Valid
Apple (AAPL)	Apple FY2022	\$394.328B	\$99.803B	0000320193-22-000108	2022-10-28	Primary 10-K Valid

4. Quality Assurance & Automated Test Results

The project includes a comprehensive 64-test automated QA suite using Pytest covering tool registries, SEC EDGAR

XML/XBRL parsers, period lineage quality gates, circuit breaker state machines, fallback managers, vector memory embeddings, calculation engines, and report synthesis. Final verification run result: 64 PASSED, 0 FAILED (100% Pass Rate).

INTERNSHIP EVALUATION SIGN-OFF

Submitted by Quaid Khan - Codebase: GitHub (Public) - Live Demo: Hugging Face Spaces (Live)

System Verification: PASSED (64/64 Unit & Regression Tests Verified)